Module 2:
Week 4: Day 1

Start Recording

What are we learning in Module 2?

- Node
- React
- APIs & Promises
- JSON Servers

Module 2 will be focused on **Frontend Development**, which means building web apps using all learned concepts

To do so, we will mainly use the frontend library **React**

Week 4 Schedule

Time	Mon	Tues	Wed	Thurs	Fri
09:00 - 09:30	Module 2 Introduction (15 mins)	Morning Review Session	Morning Review Session	Morning Review Session	Friday Activity
09:30 - 10:00	Node | Modules, NPM, Packages (75 min)	React | Lists & Keys	React | State Hook and Events	React | Routing Introduction	Morning Review Session
10:00 - 10:30					React | Forms / Controlled Components
10:30 - 11:00	React | Creating a React App, JSX & React elements				
11:00 - 11:30		React | Conditional Rendering	React | State Management	React | Routing Advanced (Params)	
11:30 - 12:00		React | Props			React | Lifting State Up
12:00 - 12:30	React | Components			React | Routing Advanced (Queries)	
12:30 - 13:00		React | Component Libraries			
13:00 - 13:30	Lunch				
13:30 - 14:00					
14:00 - 14:30	LAB | Discord Hero	LAB | Props Training	LAB | React Iron Travels	LAB | React Cohorts	LAB | React Forms
14:30 - 15:00					
15:00 - 15:30					
15:30 - 16:00					
16:00 - 16:30	Katas & Study Time	Mini Project | React App (Day 1)	Mini Project | React App (Day 2)	Mini Project | React App (Day 3)	Mini Project | React App (Day 4)
16:30 - 17:00					
17:00 - 17:30					
17:30 - 18:00					
Homework (Required)	Finish Lab HOF Katas	Finish Labs	[HOMEWORK] LAB | React State Training	Finish Labs	[HOMEWORK] LAB | React Exponents
Extra (Optional)		LAB | React Tweets	LAB | React IronContacts	LAB | React Stack Tracker	LAB | React IronNutrition

Teacher-led
Individual / Self-guided
Group / Pair Programming

What are we learning today?

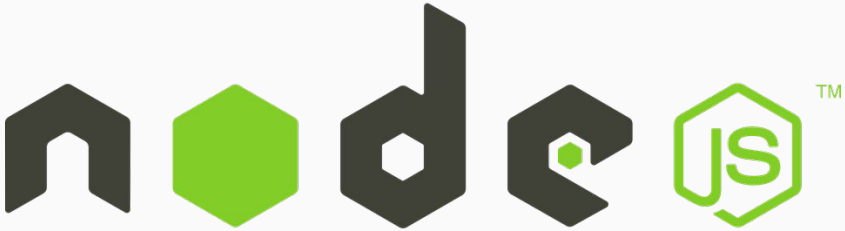
- Node Intro
- React Intro

What are we learning today?

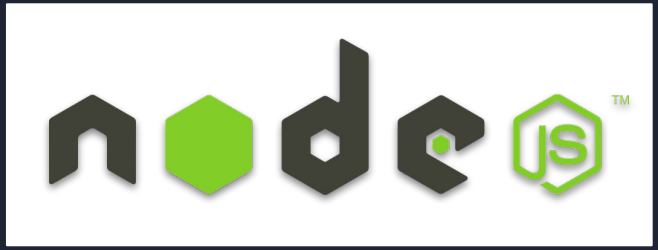
- **Node Intro**
  - **What is Node?**
- React Intro

What is node or nodejs ?

Node.js® is a **JavaScript runtime** built on Chrome's V8 JavaScript engine.



Why do we need node ?



- Can help you create a web server *(more on this on M3)*
- Can help you write server side code *(more on this on M3)*
- Can help you access/modify databases via JavaScript *(more on this on M3)*
- **Can help accessing the npm store to use external code (packages)**
- **Can help organize and run complex Frontend applications**

What are we learning today?

- **Node Intro**
  - **What is Node?**
  - **Connecting multiples files in node and JSON**
- React Intro

How to connect our files in node **ES6**

import ‘./greet.js’

Looks for greet.js file

Run all code in greet.js

greet.js

// some code to run

---

import data from ‘./greet’

Looks for greet.js file

Run all code in greet.js
Also I want your **module.exports**

greet.js

// some code to run

export default something

Break

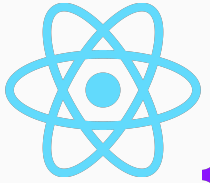
What are we learning today?

- **Node Intro**
  - **What is Node?**
  - **Connecting multiples files in node and JSON**
  - **What are packages and how do we use them**
- React Intro

What are packages ?

The term "package" is used to describe code that's been made publicly available. Generally speaking, a package helps you to add functionalities to your application.

Packages are installed locally inside our repositories via a package manager...



They sound cool, how do we use them?

What is npm ?

- Npm (node package manager) allows you to install and organize packages on your projects.
- Think of it as your App Store / Play Store where you can install all packages
- Is owned by Github & Microsoft.

npm commands

npm init -y	creates a package.json file (to be done just once per new project)
npm install PACKAGE_NAME	installs that packages and all its dependencies in your **node_modules** folder **for that project**
npm install -g PACKAGE_NAME	installs that package for **your machine**
npm install or **npm i**	Looks at the package.json file and installs all the packages listed there one by one in your **node_modules** folder

How to connect our files in node **ES6**

import data from ‘./greet’

Looks for **greet.js** file

Run all code in greet.js
Also I want your **module.exports**

greet.js

// some code to run

export default = something

---

import package from ‘greet’

Looks for installed package called **greet**

Received all code from the **package**

node_modules greet/index.js

export default = something

npm important files & folders

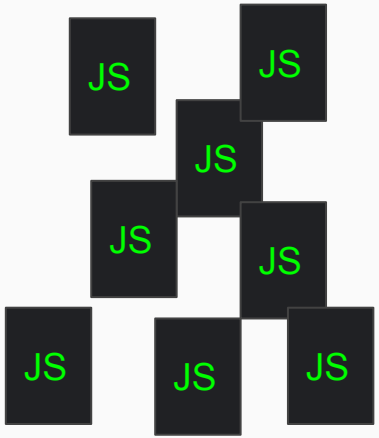
01	**package.json**	• Keeps a track of all installed packages and their versions
02	**node_modules**	• Folder where all the packages are installed when you run **npm install** or **npm i**

So many terms…

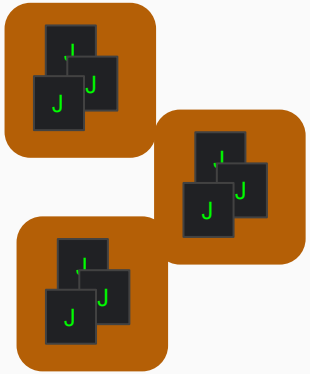
- Module
- Package
- Library
- Framework

Let’s explain them…

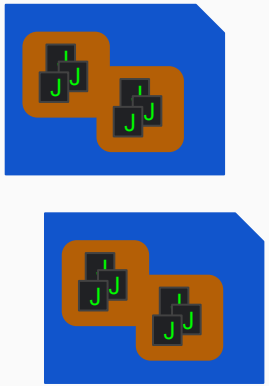
A **Module** is a file that contains various functions and variables



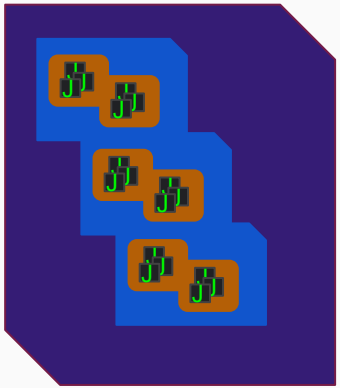
A **Package** is collection of modules.



A **Library** is a collection of packages.



A **Framework** is a collection of Libraries forming a program architecture.



What are we learning today?

- **Node Intro**
  - **What is Node?**
  - **Connecting multiples files in node and JSON**
  - **What are packages and how do we use them**
  - **gitignore file**
- React Intro

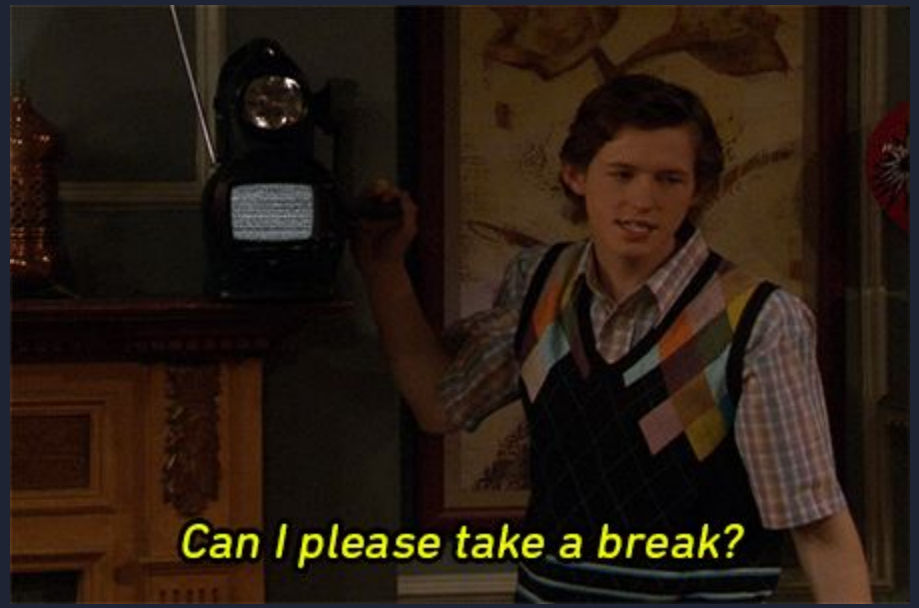
The .gitignore file

Any file name we add to it, **git** will simply ignore it for tracking purposes. It will also **not push them to github.**

We usually use this for:

- node_modules
- Secret files
- Any locally used file

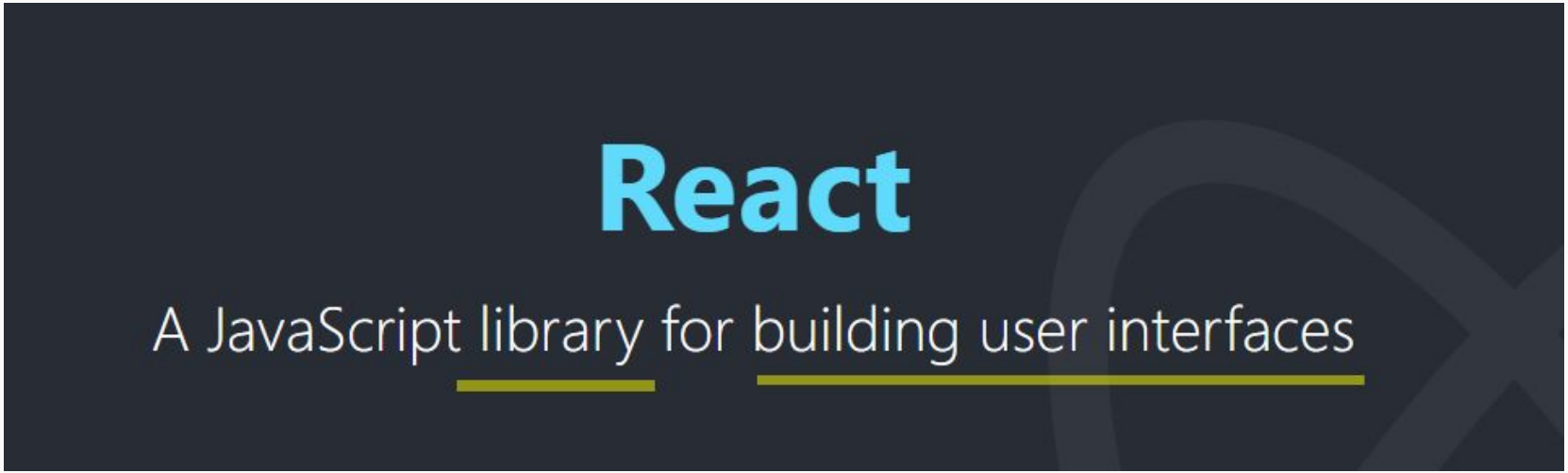
BREAK



What are we learning today?

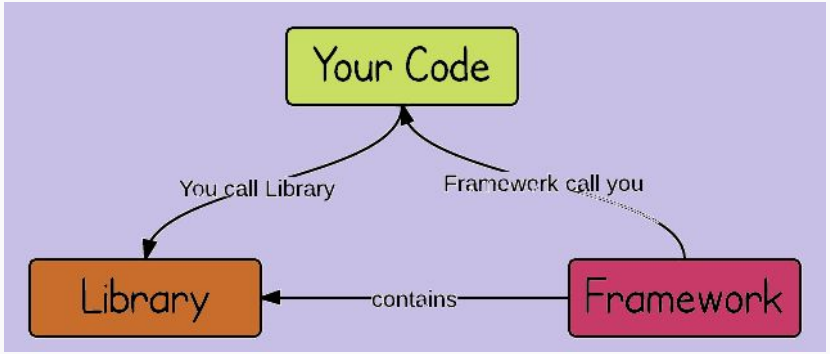
- Node Intro
  - What is Node?
  - Connecting multiples files in node and JSON
  - What are packages and how do we use them
  - gitignore file
- React Intro
  - What is React?

What is react ?



React is not a framework

Library vs Framework



	React	Angular
Components	✓	✓
Testing	Jest, Mocha	✓
HTTP library	Fetch, Axios	✓
Routing	React Router	✓
I18n	react-intl	✓
Animation	react-motion	✓
Form validation	react-forms	✓
CLI	create-react-app	angular-cli

History

2011	Created by Facebook
2012	Facebook purchased Instagram
2013	Open Sourced
2014	Facebook purchased WhatsApp
2015	React Native released
2016	React 15 was released (previous version was 0.14)
2021	The most used Frontend library
2025	Currently on version 19

Why learn ReactJS ?

What can you build with ReactJS ?

Web apps

Static sites

Mobile

Desktop

Server-rendered

Virtual Reality

Websites that use React

Desktop apps built in React (ElectronJS)

Mobile apps built in React (React-Native)

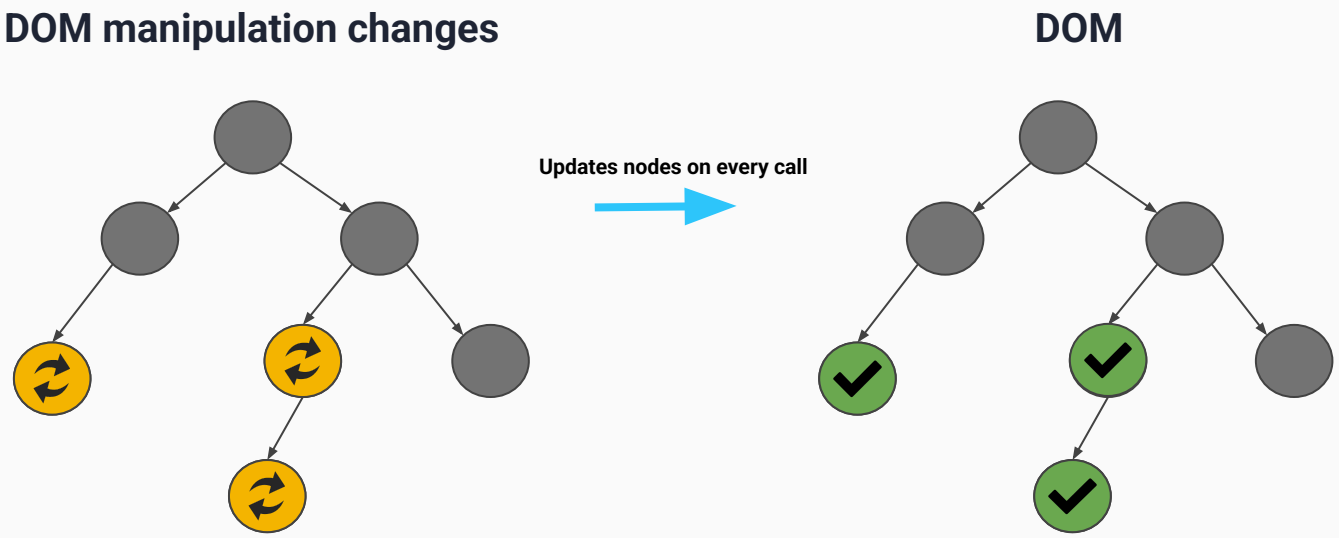
Uber Eats

Walmart

Why is React helpful ?

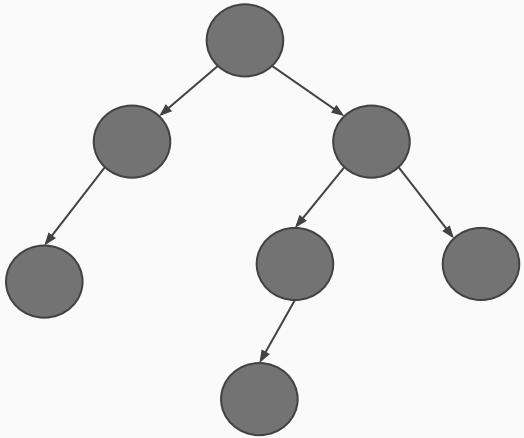
- Pages load faster than those with older ways
- The UX tools and options are drastically improved
- Easy to implement client side libraries
  - Payments
  - Maps
  - CSS Libraries
- ***Virtual DOM*** allows for faster app updates.

TRADITIONAL APP DOM

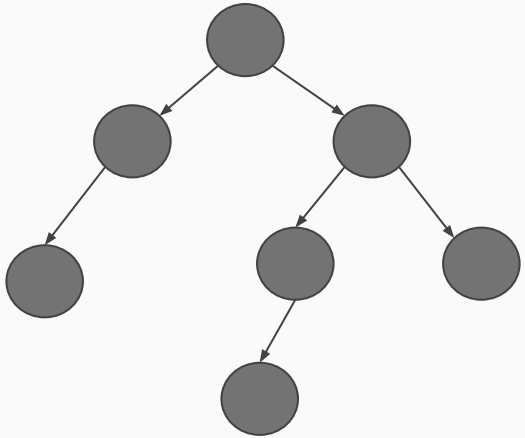


REACT APP VIRTUAL DOM

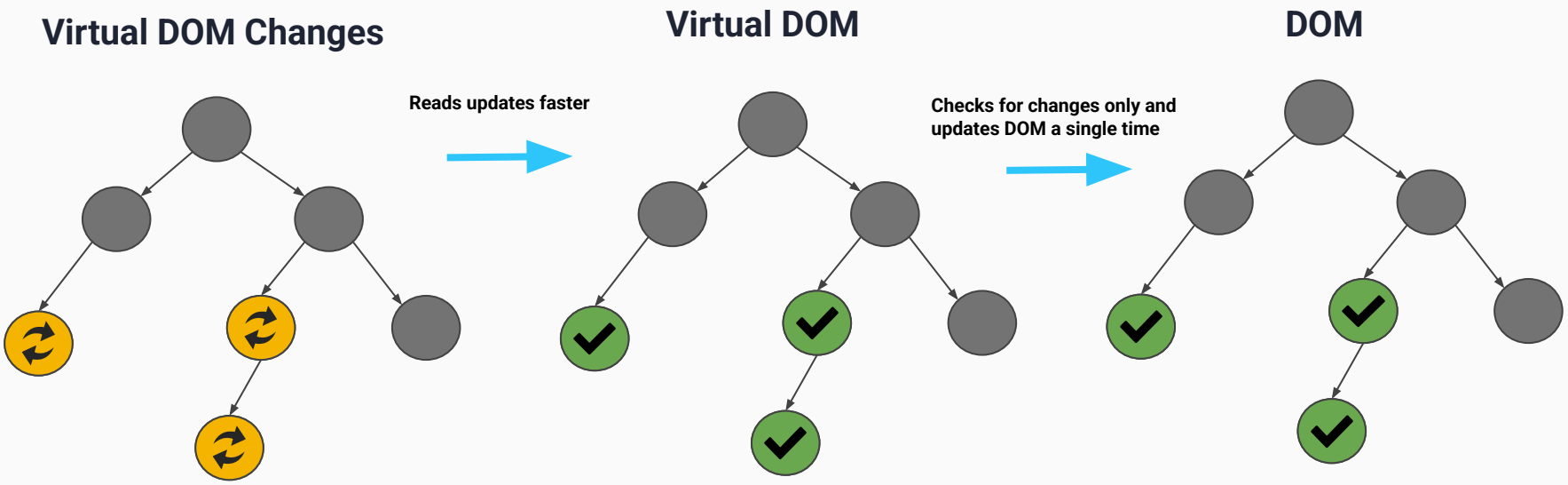
Real DOM



Virtual DOM
(Copy of real DOM that lives in JS)



REACT APP VIRTUAL DOM



Extra Resource

What are we learning today?

- Node Intro
  - What is Node?
  - Connecting multiples files in node and JSON
  - What are packages and how do we use them
  - gitignore file
- React Intro
  - What is React?
  - How to create a react app (legacy)

React main libraries

React → Knows what a **component/element** is and how to make them work together

ReactDOM → Knows how to take a **component/element** and make it show up in the DOM

Babel → Transpiles JSX code to React Virtual DOM elements.

React Legacy Setup

```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <script src="https://unpkg.com/react@18/umd/react.development.js" crossorigin></script>
 <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js" crossorigin></script>
 <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
 <title>My React Page</title>
</head>
<body>

 <div id="app"></div>

 <script src="./main.js" type="text/babel"></script>

</body>
</html>
```

React installations via html links (CDN)

Where the APP will be created/updated via Virtual DOM

Main React/Virtual DOM Script
type="text/babel" allows the transpiler to work

React Legacy Setup

main.js

```
// via CDN we have access to React and ReactDOM global variables

// Select the real DOM element where the React app will mount
const app = document.querySelector("#app")

// Create a React root to manage updates to the selected DOM node
const root = ReactDOM.createRoot(app)

// Render the React element into the real DOM
root.render(
 <h1>Hello World</h1>
)
```

What is JSX ?

+

*Similar syntax

=

```
let myName = "Bob"

let myAddress = {
 country: "Spain",
 city: "Madrid"
}
```

```
<div>
 <h2>Name: Bob</h2>
 <div>
   <h4>Country: Spain</h4>
   <p>City: Madrid</p>
 </div>
</div>
```

```
<div>
  <h2>Name: {myName}</h2>
  <div>
    <h4>Country: {myAddress.country}</h4>
    <p>City: {myAddress.city}</p>
  </div>
</div>
```

What are Components?

Components are functions that return JSX code to be rendered by the React Virtual DOM

React Components

Component Type	Creating Components	Invoking Components
Functional Component	`function Hello() { // ... any JS code logic return <div>Hey! React is awesome!</div> }`	`<Hello />`

NOTE THIS DOWN

- **Components** can be reused in many places in your code
- **Components** must be named in **Pascal Case**
- **Components** will always return JSX (to render the info)
- **Components** are invoked using **<Name />** and never Name()

- Everything in react is **JSX** , not **HTML**

As React is only a library it depends on other tools that provide a development environment.

As React is only a library it depends on other tools that provide a development environment.

- Package Manager
- React Packages
- React Dependencies
- Compiler (Babel)
- Bundler (Webpack or Parcel)

And everytime we want to start a new project we don't want to setup everything

What are we learning today?

- Node Intro
  - What is Node?
  - Connecting multiples files in node and JSON
  - What are packages and how do we use them
  - gitignore file
- React Intro
  - What is React?
  - How to create a react app (legacy)
  - How to create a react app (with VITE)

React additional tools

VITE

VITE is a **generator** that serves as a boilerplate for different types of Frontend Libraries and Frameworks (including REACT).

It includes all basic environment packages needed as well as an initial structure

npm create vite@latest	*To create the React Boilerplate*
cd project-name	*To access the created folder*
code .	*To open VS code*

Then either in the main terminal or inside the VS code terminal…

npm install	*To install all dependencies*
npm run dev	*To run the React App*

What are we learning today?

- Node Intro
  - What is Node?
  - Connecting multiples files in node and JSON
  - What are packages and how do we use them
  - gitignore file
- React Intro
  - What is React?
  - How to create a react app (legacy)
  - How to create a react app (with VITE)
  - Component Based Structure

What is a Component based structure ?

Search

Olenna | Home | Find Friends

Olenna Mason

News Feed
Messenger
Watch
Marketplace

Shortcuts
FarmVille 2

Explore
Events
Pages
Groups
Friend Lists
Fundraisers
See More...

Create
Ad · Page · Group · Event · Fundraiser

Make Post | Photo/Video Album | Live Video

What's on your mind, Olenna?

List | Photo/Video | Tag Friends

Julia Fillory is in Raleigh, North Carolina.
2 hrs

Checking out some new stores downtown!

Raleigh, North Carolina
City · United States
Bryan Durand and 2 others have been here

Save

Stories | Archive · Settings

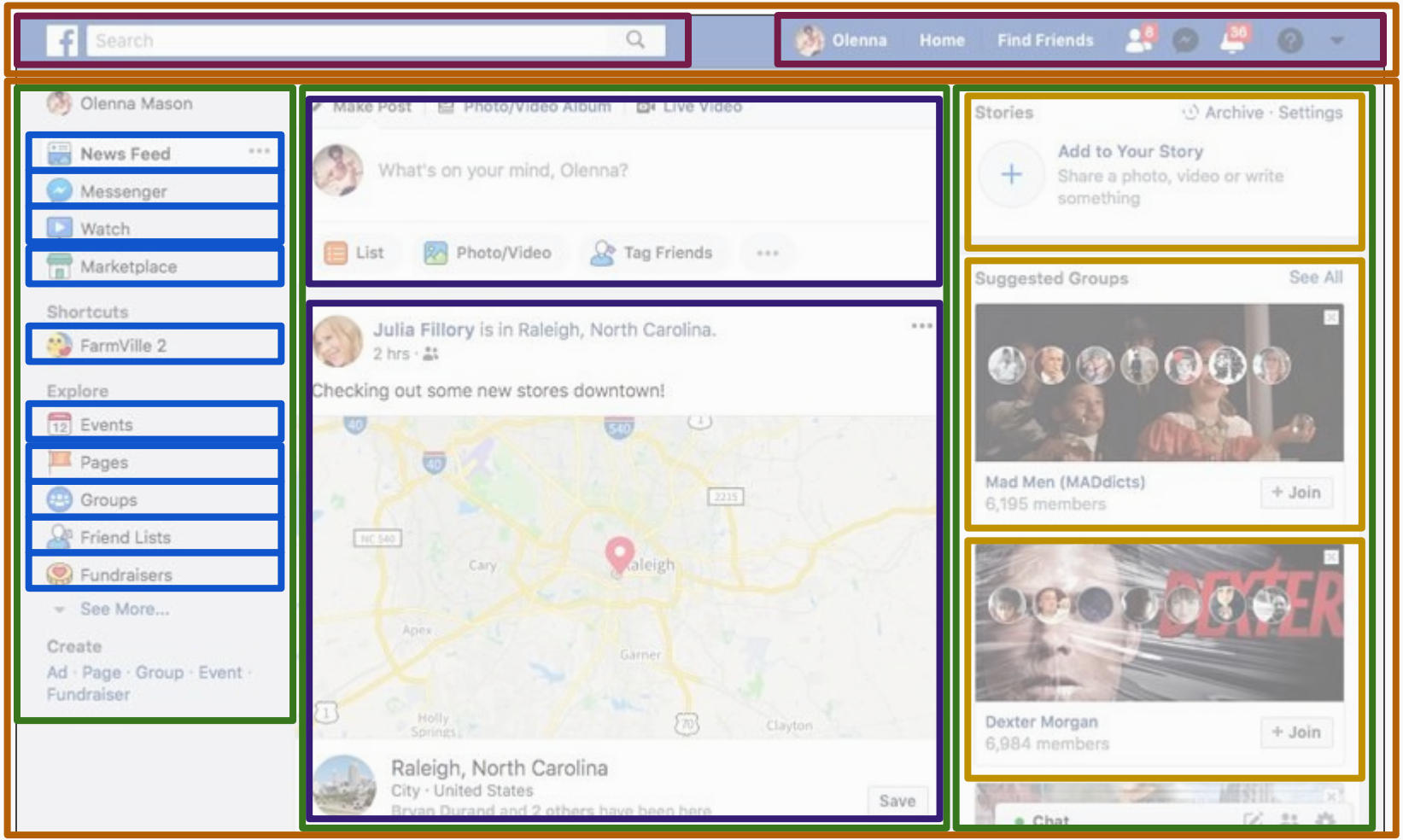
Add to Your Story
Share a photo, video or write something

Suggested Groups | See All

Mad Men (MADdicts)
6,195 members
+ Join

Dexter Morgan
6,984 members
+ Join

Chat


Search
Olenna
Home
Find Friends
Olenna Mason
News Feed
Messenger
Watch
Marketplace
Shortcuts
FarmVille 2
Explore
Events
Pages
Groups
Friend Lists
Fundraisers
See More...
Create
Ad · Page · Group · Event · Fundraiser
Make Post
Photo/Video Album
Live Video
What's on your mind, Olenna?
List
Photo/Video
Tag Friends
Julia Fillory is in Raleigh, North Carolina.
2 hrs
Checking out some new stores downtown!
Cary
Raleigh
Apex
Garner
Holly Springs
Clayton
Raleigh, North Carolina
City · United States
Save
Stories
Archive · Settings
Add to Your Story
Share a photo, video or write something
Suggested Groups
See All
Mad Men (MADdicts)
6,195 members
+ Join
Dexter Morgan
6,984 members
+ Join
Chat

Doing inline style in React

```
function Card() {

  const style = {
    backgroundColor: "darkblue",
    width: "200px",
    color: "white",
    fontSize: "20px",
    padding: "5px"
  }

  return (
    <div style={style}>
      <h1>Hello</h1>
      <h1>World!</h1>
    </div>
  )
}
```

IMPORTANT!

- Style property always expects an object with all style attributes
- Style attributes are all in camelcase
  - **NO** Background-color
  - **YES** backgroundColor

Installation time!

React Developer Tools

What did we learned today ?

- Node
  - What is node
  - How to move things around in node (import)
  - What is npm and packages
- What is **React**, **JSX** and the **Virtual DOM**
  - How to create a **React App with Vite**
  - What are **Components** and their types
  - **How to do inline styles**

Deliverables & Homework

- React Discord Hero Lab
- Adv JS topics activities (next slide)
- HOF Katas (map, filter, reduce, etc…) (see student portal)

Activities from last week

Activity I - Destructuring

Activity II - Spread & Rest

Activity III - Advanced JS topics